

Three-dimensional fascia-oriented training

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Introduction

Fascia is a ubiquitous, three-dimensionally networked structure, which takes on different forms depending on the nature of the load present (Stecco et al., 2007; Schleip et al., 2012). This varying structure is certainly not “in-born”, rather it is formed by the individual strains on the body, according to the motor development of a person (Blasi et al., 2015; Wilke et al., 2016). The developmental steps one can observe usually occur in a pattern: from gross movement towards a target action, from experimental movement to movement pattern, and from repeated prevention of falling to locomotion. On confronting gravity, the exclusively kyphotic curve that is characteristic of the spine of a newborn baby is transformed, taking on a S-shaped curve that extends along the sagittal plane. The baby’s crawling motion first produces lateral flexion and then spinal rotation as a function of that side-to-side movement. During this time, the movements are driven entirely by neuromuscular activity as the myofascial reinforcement present in an adult body is as yet undeveloped (Schultz et al., 1996). Only through repeated use and the related tensile loading does the tissue become more dense and stronger over months and years and, depending on the level of physical activity and movement habits, integrated into the individual’s movement dynamics.

Strong differences can arise, for example, between people who suffer with paralysis from

birth and those who are exposed to a very high level of repetitive or one-sided strain: for the person in a wheelchair, the fascia lata is not palpable as a reinforced structure whereas in athletes it is well-defined; for cowboys who spend hours every day on the back of a horse, a corresponding reinforcement of tissue can be found on the insides of the upper legs (Otsuka et al., 2018).

Myofascial coordination and fascial trigger mechanisms

Fascial reinforcement can serve to lessen strain on the musculature because it takes on a large portion of the load in eccentric (Chapter 26) as well as slow or almost static movements. It, furthermore, shapes the posture of the body via its base level of tension. The harmonic interplay between fascia and the involved musculature is thus a characteristic of a properly functioning system (Holleran et al., 1995). If the body perceives pain, then this interplay can be impaired. This leads to an increase in persistent tension of the musculature, which contributes to acidosis, fibrosis and fascial adhesions, and if not resolved the pathology could become chronic (Pavan et al., 2014).

It would be sensible here to keep the concept of myofascial coordination in mind because the muscular system and the fascial system are obviously inseparably linked, and each is dependent on the functionality of the other.

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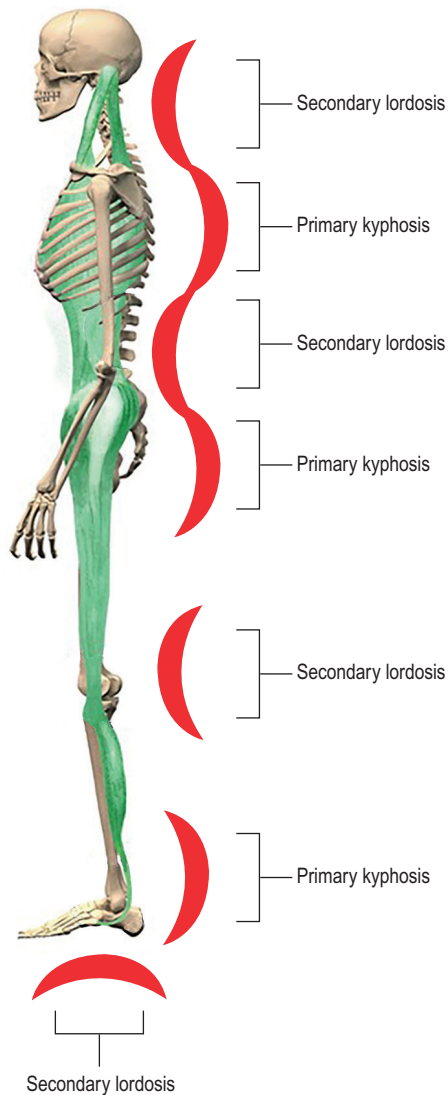


Figure 40.1

The lordotic curves of the body result first from muscle tension. Then they become reinforced by fascial structures.

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This connection becomes particularly obvious for quick or elastic movements. Here, it is not

about a gradual transfer of tension, rather the fascial regaining force is employed; the Achilles tendon is one example. For repeated two-legged hopping with brief ground contact, it appears that a large amount of the energy for the jump can be gained by the stretching of the Achilles tendon (Kawakami et al., 2002). During that movement the calf musculature only has an isometric function: it has to bring the Achilles tendon to the right level tension in anticipation for jumping, so as to generate its “catapult effect” (Sugisaki et al., 2011). Repeated studies in athletes have shown that the interplay between the fascia and muscular pre-tensioning is trainable (Fouré et al., 2012; Hirayama et al., 2017) and apparently this is also the case for the increasingly elastic movement quality of athletes in most sports.

Aside from these parallel and serial mechanisms of muscular and fascial tension, a spatial deformation of the muscle can be observed as it increases in thickness with contraction, and this then deforms the epimysium which transmits power to adjacent structures (Findley et al., 2015).

Tensegrity

Fascia training is thus also always muscle training, and muscle training also always affects—in varying amounts—the fascial system. The more naturally, gracefully or explosively a movement is executed, the more the fascial tissue becomes involved and trained. If one accepts the tensegrity model as the basis for human posture and movement, two factors emerge: the joint-stabilizing function of the myofascial base tension, and long-chain impulse transmission along the myofascial lines of tension.

According to the model of tensegrity, human bones function like distance pieces, kept in place by tension within the long lines of myofascial

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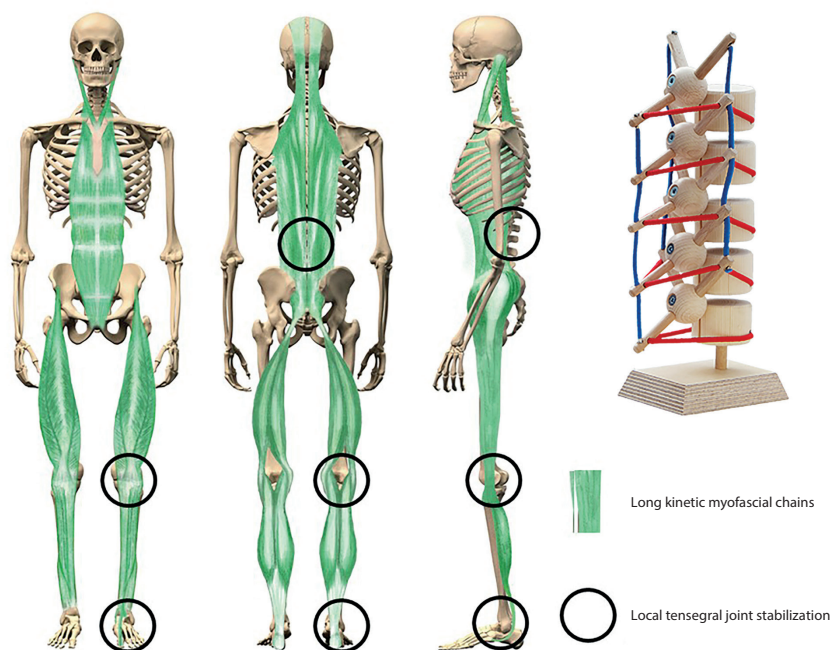


Figure 40.2

Tensegrity: local joint stabilization and long myofascial chains.

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connections. The position of the bones is dictated by means of the tensile elements, and the bones somehow “float” in a fascial frame of tension (Dischiavi et al., 2018). This is why a harmonic balance in tension guarantees functional posture and movement.

An interesting side effect of this model is that, given these conditions, compressive stress does not exist in the joint because the joints are actually not needed as contact surfaces. By contrast, the positive tension surrounding the joint, i.e. tensegrity, even leads to less strain on the joint. While in the traditional and common model of linear-functioning muscle tension, where each increase in tension leads to an increase in load on the joint, targeted tensioning allows for and delivers relief for the joint surfaces through a three-dimensional, spiral and oblique arrangement of tension around the joint (Hashizume et al., 2014; Dischiavi et al., 2018).

Training of the tensional integrity system is markedly more difficult than common strength training, as it requires more attentiveness, acuity and body awareness. Such exercises can be found in existing body and movement schools, even if their principles are not explicitly formulated as being fascial in nature. These include approaches to the resolution of existing dysfunctional tension with Eutony (Alexander, 1983), the Feldenkrais Method and the Alexander Technique. Here, the training mostly takes place through inductive movement imagery (ideokinesis), which serves to coordinate the structure of individual tension elements. To this end, some movement schools use movements that are primitive in origin, for example, Continuum Movement or Animal Moves, which elaborate on and imitate the movements of various animals. Others, like Crawling Fitness, seek to trace human movement development by creating a

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modern and improved form of Klapp's Crawling. Still others look to kinematic concepts of strength and tensions centers (Pilates) or lines of tension (Spiraldynamik, Gyrokinesis). The orientation of movement in space represents a further element, which can be found in Laban Movement Analysis or the Gyrotonic Method.

Fundamentals: the three-dimensional architecture of the foot

If one adheres to the model of Spiraldynamik then the arch of the human foot is not determined by the shape of bones of the feet or the linear tension of the foot musculature, rather it



Figure 40.3

Three-dimensional training in Functional Fitness.

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Figure 40.4

Three-dimensional training with a special apparatus: Gyrotonic Method®.

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results from the spiral screw-like connection of the intricate forefoot in relation to the solid form of the heel. While great apes possess a similar boney foot structure, this spiral screwing action is missing; rather, they primarily put weight through the outside edge of the foot, which is why they run stiffly on their back legs (Gray, 2016). Whenever people attempt to simulate a foot arch in this way, exactly the same precarious and unstable gorilla-like gait is the result.

In order to achieve a healthy arch in the structure of the foot, a number of things need to be done:

- Structural work on the fixations and thickening of the connective tissue of the foot, particularly the soles of the feet, the most painful point under the tarsal bones (navicular), the forefoot bones (metatarsals) and the area around the ankles (retinaculi). This can be done manually, or possibly using tennis or foam balls. Furthermore, initiation of the spiral screwing can be achieved through manual work (Figure 40.5).
- Bring awareness to the three contact surfaces when standing: the heel, the ball of

the foot proximal to the little toe, and the ball of the big toe. When actively loading the ball of the big toe, the heel must remain stable. Neither the foot nor the knee should give way or fold inwards for this action.

- While maintaining an emphasis on the loading of the ball of the big toe, the stable heels can be raised a bit and, if a step is available, the heels can also be lowered.
- If a sufficient level of coordination is present, the loading of the big toe can be practiced in movement: first simply by shifting weight forward and backward, then in walking backwards, and finally in forward locomotion.

Stabilization: three-dimensional body tensioning

Almost without exception, the joint partners in human anatomy are comprised of surfaces that fit more or less well on top of one another, which must be held in their functional position through appropriate muscular, ligamentous and capsular tension. As mentioned above, the oblique



Figure 40.5

Structural work.

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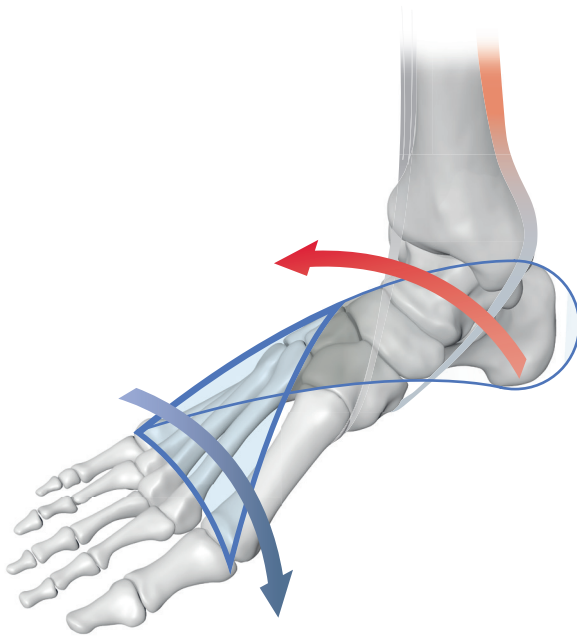


Figure 40.6

Spiraling model of the foot.

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and spiral arrangement of the tension elements makes such stabilization possible. Again we find this model in Spiraldynamik, in the Franklin® Method known as “bone rhythms”, in the Gyrotonic Method as “narrowing” (in dance), as “expanding”, or also as “powerhouse” in the Pilates method. In all cases, the body seems to become taller or longer, and at the same time well-balanced and more stable.

An absence of such tension can be observed in people who are standing a lot every day, and shifting their body weight primarily to one side. In doing this, the loaded leg seems to screw inwards, the arch of the foot collapses, the lower leg rotates to the outside, the upper leg rotates to the inside and the knee is in turn locked in place.

This position does indeed require a minimum of muscle tension, but puts a considerable strain on the joints. In order to reverse this sinking

inwards, the muscles that move the joint must be active and the joint partners have to de-rotate against gravity. The forefoot must push against the ground via the ball of foot behind the big toe and the heels have to remain stable; while the lower legs rotate internally, the upper legs rotate externally and the ilia again rotate inwards. One can imagine pushing with the feet and legs against a floor that yields to the force, or similarly the lengthening action of the legs as a downward moving elevator comes to a stop and gravity is thus enhanced.

This image of enhanced gravity or tension-induced lengthening is then transferred to the torso and the upper extremities. While the arms function in a similar way to the legs, the torso has at its disposal a number of additional oblique and spiral connections, and moreover additional horizontal tension through the transverse abdominal muscles, pelvic floor and diaphragm.

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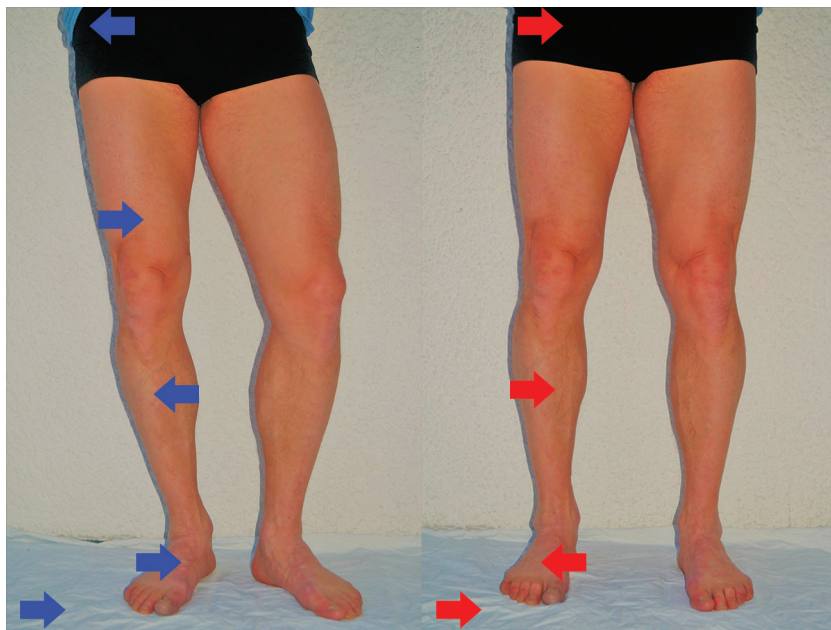


Figure 40.7

Compressed and decompressed joints via active derotation. www.faszienfit.de.



Figure 40.8

Gyrotonic® an Gyrokinesis® as an example of elongation/expansion).

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The most graceful combination of simultaneous three-dimensional stabilizing and expansive movement is clear to see in the Gyrotonic Method.

Training required: expansion in motion

A striking characteristic of the pre-tensioning mentioned above is that the amplitude of

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Figure 40.9

Gyrotonic® an Gyrokinesis® as an example of elongation/expansion).

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movement seems to be reduced in the first moment. This is obvious, because a movement using maximum range of motion requires “long”, i.e. relaxed, muscles. But through the initiation of pre-tensioning, the elastic forces around the joint become active earlier. In this respect, two advantages of pre-tensioning come together: on the one hand, the joint is kept safe and the joint surfaces are relieved of strain, and on the other hand, elastic power is available for rapid movement.

Since the pull of muscles does not only act in a linear way on the skeleton, but its forces are distributed to the capsular and ligamentous structures around the joint, the linear power available for a movement is reduced by 20–30% (Maas et al., 2001). In order to develop a movement that is more refined and easier on the joints, the focus should be shifted an even greater amount in favor of the capsule tension so that the person performing the movement might be only making 50% or less power available for the target movement. For practice, to whatever extent is possible, the main portion of the power should contribute to the positioning of the joint. For all practical

purposes this is achieved through kinetic images of great resistance (e.g. weight, pull, water), against which a movement is made. Further useful ideas include the imaginary extension of a part of the body that participates in the movement or a yawning quality of movement (pandiculation). The base tension is always at least as important as the target movement. Narrowing, expanding or the stable powerhouse are movements done with this in mind.

The initial over-tensioning on the joint capsule certainly can (and must!) be increasingly reduced in advanced practitioners until the movement is made in a truly elastically way, although the body does not lose length and the movement does not lose amplitude. A well-coordinated body, in this sense, anticipates the coming load and establishes just the right amount of pre-tension.

Long connections: myofascial functional chains

One of the most well-known myofascial functional chains is the long superficial backlines.

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The fascial connections are really three-dimensional, but with spiral and spirally functional lines. These have been amply established in research (Wilke et al., 2016) and connect the torso with the respective contralateral extremities. Any locomotion, jumping or throwing motion requires this crossing-over connection. Whether in soccer, golf or javelin throwing, as soon as an object needs to be moved, the respective movement of the extremities is not only supported by the torso. An optimal movement impulse is also carried out through the spiral pre-stretching and successive shortening of the myofascial tracts.

Comparing the throwing abilities of gorillas and people, the apparent lack of this spiral-screwing motion is striking in the great apes, a characteristic limitation in contrast to the human thrower (Goldman, 2014). Great apes accelerate an object like a person who is just learning to throw, and moreover has not practiced very much. It is solely an arm movement: the monkey has anatomical limits to movement and the human beginner who has not yet learned to coordinate the movement are thus

similar throwers. In order to really utilize the elastic energy of the contralateral-oblique myofascial tracts, differentiated timing is required in terms of the muscular tensioning in the eccentric lengthening phase, isometric support during the maximal stretch and the elastic power of the fascia in the acceleration phase.

In order to train this quality of movement, the “reaching back” aspect has to be clearly emphasized. This involves keeping both the amplitude and also the source of the movement impulse in mind. In the field of strength training, this is described as dynamic eccentric or plyometric training. At the turnaround point of the movement, when the “reaching back” is transferred into the concentric target movement, maximal tensile forces are realized. If this movement coupling happens quickly enough, the body can utilize the elasticity of the tendons and fascia and maximally accelerate the body or object, often called the “catapult effect” (Kawakami et al., 2002). The timespan for this is dependent on things like the length of the myofascial chain: the longer the elastic chain, the longer the reversal



Figure 40.10

Throwing movement/
stretching.

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of motion can last. For example, while skipping rope with a low jump height, the ground contact time is under 0.1 s; it amounts to about 0.15 s for a long jump and, for a high jump, for which the entire body is elastically involved, about 0.2 s. The optimal value of the duration of this elastic movement reversal is different among individuals and depends on the environment, but it decreases in general as progress is made in athletic training, which reflects the increasing role of the fascial elasticity (Fouré et al., 2012; Hirayama et al., 2017).

Three-dimensional equipment training

The most ineffective forms of movement, from a functional as well as a fascial standpoint, involve isolated, limited, single-joint loading, like those that are dictated by some strength training equipment. A knee curl machine only offers a movement that is neither present in any sports nor any everyday activity. Such training only makes sense for countering muscular imbalance, or if one is working with people who are absolute beginners when it comes to physical activity. With increasing athletic experience, the equipment training must become more complex,

as fascia connects structures in multiple directions, involving more joints, i.e. closer to natural everyday movements, because of its elastic storage function. Newer trends, like Functional Training, Animal Moves or Paleo Training shift the person training into a natural wilderness, to work against these conditions. The delight of self-experience through proprioception (which derives from fascial tissue) in these archaic forms of movement has a balancing effect, making them restorative and enhancing creativity to compensate for a lack of everyday movement.

Concepts like Pilates or the Gyrotonic Method place emphasis on improving the control of movement, and thus position fascia-related whole-body coordination in the foreground. The devices of both of these disciplines are expressions of this attempt to support the manifold human movement possibilities with appropriate equipment.

Nonetheless, classic equipment like steel cables, free weights or other systems also allow for functional application: it is possible to use bodyweight training, and in so doing attempt to maximally train the possibility for differentiated movement.

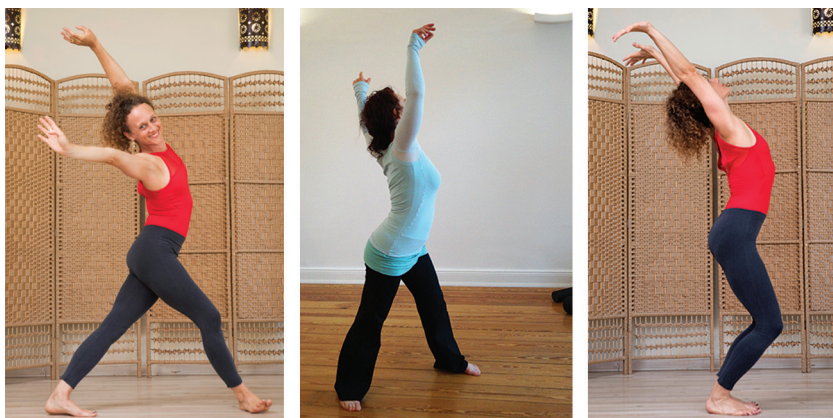


Figure 40.11

Dancing workout with expansion in all dimensions.
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Conclusion

The modern world of everyday movement is becoming increasingly two-dimensional. Many people sit in front of screens for a long time, and as a result fall into a more and more hunched posture. This tendency towards imbalance leads first to habitual postural adaptations and then to structural adaptations of the soft tissue, which in turn cause imbalances, overloading syndromes and degenerative processes (Bishop et al., 2014; Pavan et al., 2014). In the past, advice

included “sit up straight”, which fortunately has been replaced with “sit movable”. Workplaces are being designed to be more adaptable and more “movement friendly”. Nevertheless, the crucial and decisive factor is the movement experience of the individual themselves. In athletic activity as well as in therapy and everyday movements, there is a need to carry out complex and three-dimensional movement tasks, resulting first in the enhancement of neuromuscular coordination and subsequently the improvement myofascial structures.

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